

E-Commerce Marketplace Analysis – Technical Assessment

Scenario

Your company operates an online marketplace. You have received a [sales dataset](#) containing customer transactions, product details, and order information.

Your task is to analyze key business metrics, identify performance issues, and provide **actionable, data-driven recommendations** to improve marketplace performance.

Dataset Columns

Column Name	Description
Row ID	Identifier for the row
Order ID	Unique order identifier
Order Date	Date when the order was placed
Ship Date	Date when the order was shipped
Ship Mode	Shipping method (e.g., Standard Class, Second Class)
Customer ID	Unique identifier for the customer
Customer Name	Name of the customer
Segment	Customer segment (Consumer, Corporate, etc.)
Country / Region	Customer country or region
City	Customer city
Postal Code	Customer postal code
Region	Geographic region (e.g., East, West)
Product ID	Unique identifier for the product
Category	Product category (Furniture, Technology, etc.)
Sub-Category	Product sub-category (e.g., Bookcases, Labels)

Product Name	Name of the product
Sales	Total sales amount
Quantity	Number of units sold
Discount	Discount applied
Profit	Profit generated from the sale

Assessment Tasks

Part A — Data Preparation

1. Load the dataset into **SQL**, **Python (pandas)**, or any analytics tool of your choice.
 2. Validate data quality:
 - Missing values
 - Duplicates
 - Invalid or inconsistent values
 3. Create derived features and flags:
 - **Active vs Inactive Customers**
(e.g., based on purchases in the last 90 days)
 - **Repeat Customers**
(customers with more than one order)
 - **New Customers**
(first-time buyers in the current period)-- Monthly
 - **Churned Customers**
(customers who purchased in a previous period but not in the current period)-- Monthly
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Part B — Business KPIs

Calculate and clearly define the following metrics:

- **Active Customers**
Customers who made purchases during the selected period.--Monthly
- **New Customers**
Customers whose first purchase occurred in the selected period.-- Monthly
- **Churn Rate**
Percentage of customers who did not return compared to the previous period.--Monthly
- **Retention Rate**
Percentage of customers who made repeat purchases.
- **Average Order Value (AOV)**
 $\text{Total Sales} \div \text{Number of Orders}$
- **Customer Lifetime Value (LTV)**
Average revenue generated per customer over time.
- **Product & Category Performance**
 - Top products
 - Top categories and sub-categories
- **Customer Segmentation**
 - RFM analysis **OR**
 - Grouping by frequency and monetary value
 - *(Optional)* Use machine learning clustering techniques

★ Bonus Task

- **Churn Prediction Model**
Build a machine learning model to classify or predict churned customers and explain:

- Feature selection
 - Model choice
 - Evaluation metrics
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Part C — Insights & Visualization

Build dashboards or visualizations that include:

- Customer cohorts and **retention curves**
- Top products, categories, and sub-categories
- Sales trends over time (**monthly / weekly**)
- Shipping mode adoption (if relevant)
- Regional or city-level performance
- Profitability trends

Tools:

Metabase, Power BI, Tableau, or Python visualization libraries (matplotlib, seaborn, plotly)

Part D — Business Recommendations

Based on your analysis, provide clear and actionable recommendations:

- Why are some customers churning? How can churn be reduced?
- Which product categories or sub-categories should be promoted?
- What pricing, bundling, or discount strategies could increase **AOV** and **profitability**?
- Which regions or customer segments need attention?

- How can shipping or delivery be improved to enhance customer satisfaction?

Deliverables

- **SQL queries and/or Python notebooks** showing all calculations
- **Dashboards or visualizations** for key KPIs
- **Executive summary / report** presenting insights and business recommendations